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Cushion with Independent Air Chambers integrated with a Pressure-Sensitive MAT

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Abstract

Athletes with spinal cord injuries (SCI) using Functional Electrical Stimulation (FES)-cycling face significant risks of **discomfort and pressure ulcers during training and competition**, due to prolonged sitting and impaired postural adjustments. Based on feedback from FES-cycling athletes and inspired by their need for improved comfort and safety, we designed an **innovative pressure management system integrated into the trike seat**. The solution consists of a custom-made cushion with 18 independent air chambers dynamically controlled by a microcontroller, and two high-resolution pressure-sensitive mats (*Tekscan CONFORMat™*) that monitor the pressure distribution in real time.

The system continuously evaluates pressure at the user-seat interface. When excessive pressure is detected in critical areas for a prolonged time, the system selectively deflates the corresponding **air cells**, redistributing the load and reducing the risk of pressure ulcers. A microporous polyurethane **cover** ensures breathability, waterproofing, and user comfort, while maintaining hygiene and durability. The embedded electronic system, managed by an **Arduino Mega 2560**, interprets sensor data and actuates a distributed **electropneumatic network** including valves and a compressor. The software integrates a pressure-time risk model based on the Reswick and Rogers curve to trigger protective actions.

The design is modular, lightweight, and adaptable to existing trike configurations, offering enhanced comfort, safety, and potentially better athletic performance. Despite current limitations such as wiring complexity and overall weight, this prototype demonstrates a strong potential for real-world implementation and further development. Ultimately, the project represents a significant advancement in integrating assistive technologies for **rehabilitation and competitive sports**, supporting athletes with SCI in pursuing their goals safely and comfortably.

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1. Introduction

According to the World Health Organization, over 15 million people worldwide live with a **spinal cord injury (SCI)**. In recent years, thanks to the spread of the “Sport-Therapy” model, it has been shown that, after an SCI, physical training can help improve a person’s physical and psychological well-being through the use of Functional Electrical Stimulation (FES) during cycling exercises (**FES-cycling**) [1,2].

FES is a type of electrical stimulation combined with a functional task: it activates the nerves that control the muscles affected by paralysis or neurological disabilities, promoting muscle contraction. In FES-cycling, a trike—a three-wheeled bicycle adapted for people with motor impairments—is used. This technology has gained visibility through events like the Cybathlon, an international competition for athletes with disabilities who use advanced assistive devices.

The trike used in our project is based on a modified commercial recumbent tricycle (CATRIKE 700). It includes instrumented pedals to measure forces, an ankle-foot orthosis (AFO) to stabilize the ankle and restrict movement to the sagittal plane, and a 24V mid-drive DC motor with freewheel, controlled via PWM and an H-bridge. Pedaling is monitored using a magnetic encoder, while a neuromuscular stimulation system activates up to eight muscles through electrodes. The entire setup is managed by a Raspberry Pi 4, with a user interface on an Android tablet and physical push buttons for control. All components are integrated behind the seat for easier transport. The system can be powered by battery or through a wall outlet [3].

Despite the progress made, using the trike still requires further development to improve the athlete’s overall experience in terms of comfort, safety, and performance.

2. User-center need and project proposal

From what was said during the [focus group](#), the athletes clearly expressed a shared need: improving both comfort and safety while using the trike. Comfort is connected to having a **more ergonomic seat**, while safety is mostly related to **preventing pressure sores**, which is their biggest concern.

In healthy people, discomfort is reduced by making small movements while sitting, called In Chair Movements (ICM) [4,5]. These are small posture changes that help relieve pressure on certain body areas. But for people with disabilities, this is not possible. In our case, because of the athletes’ condition, they cannot make these movements. In everyday situations, they often use their arms to lift themselves slightly off the seat for a few seconds to reduce pressure. However, during training or races, this is not an option. Because of this, they are at risk of developing injuries that can turn into pressure sores. These sores are caused by sitting in the same position for too long, which reduces blood flow and damages the skin and tissues. They usually appear in areas where there is a lot of pressure, like the sacrum, buttocks, and thighs. If not treated early, they can lead to serious complications.

To prevent **pressure sores**, there are usually three main strategies: changing position often, sitting with good posture, and using support devices like anti-decubitus cushions. But during sports activity, it is very hard to follow these strategies, so we need smarter and more effective solutions.

Our project aims to solve this problem, so that athletes can use the trike in a safer and more comfortable way, without constantly worrying about pressure sores. These injuries are often discovered too late and can cause even more problems in everyday life. To face this issue, it is very important to study how pressure is distributed between the person and the seat. Understanding this will help us improve both **safety and comfort**, and maybe even help the athletes perform better in the future.

To solve these problems, we have identified a list of important requirements that our device will need to meet.

2.1 Functional requirements

The **functional requirements** are related to the behaviour and performance of the system and are essential for the correct use of the device. Therefore, the product must:

- **prevent pressure sores** by redistributing pressure effectively, avoiding long-lasting pressure in critical areas;
- **ensure user comfort** by making the seat comfortable even during long periods of use, through the use of optimal shapes and materials;
- **maintain postural stability** by preventing unwanted movements, both of the user (such as sliding sideways or forward) and of the system itself with respect to the trike, by including non-slip surfaces and stable, secure fixing mechanisms;
- **be adaptable to the trike** so it can be easily integrated with its structure;
- **be waterproof and resistant** to keep working well even in difficult environmental conditions (sweat, humidity, heat, accidental impacts).

2.2 Non-functional requirements

The **non-functional requirements** refer to aspects that, even if they do not directly affect how the system works, are still important to make the product effective and ensure a good user experience. So, the device also needs to be:

- **not bulky and lightweight** so it does not affect performance during races;
- **easy to carry** so it can be moved from one trike to another both during competitions and research phases;
- **easy to use**, so it can be assembled, removed, and adjusted either by the athlete or by an assistant;
- **simple to maintain**, with components that are easy to reach for checks, cleaning, or replacement, making the system long-lasting and practical.

3. Project design

Our work resulted in the development of an innovative system for dynamic pressure management: **a cushion with independent air chambers integrated with two pressure-sensitive MATs**. The goal of this solution is to optimize weight distribution, prevent pressure injuries, and improve user comfort through an intelligent and adaptive system.

The cushion with independent air chambers is designed to change its structure in response to pressure changes detected by the MATs, which collect real-time data about weight distribution. When a certain area is under too much pressure for a long time, the system reacts by adjusting the amount of air in the corresponding chambers, reducing overload and improving overall weight balance.

The device is made up of:

- **cover**: the top layer of the system, directly in contact with the user, chosen to ensure comfort, hygiene, and safety;

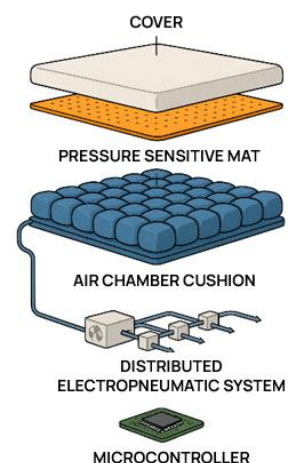


Figure 1 - Project design structure

- **pressure-sensitive MAT:** a sensor mat integrated into the system, placed between the cushion and the upper cover;
- **cushion with independent air chambers:** a cushion made of several separate air cells, located at the bottom of the system;
- **distributed electropneumatic system:** a set of valves and a compressor for dynamic control of the cushion's air pressure, placed underneath the cushion;
- **microcontroller:** the integrated circuit that acts as the brain of the system, receiving data from the MAT and controlling the action of the valves and compressor.

3.1 Cover

The choice of material for the top cover is strategic, as it is the direct interface with the user.

Polyurethane (PU) is considered the ideal solution for the cover of rehabilitative devices. Taking all parameters into account, a **microporous PU cover** with a thickness of 150–250 μm offers the best compromise between comfort, breathability, resistance, and hygiene, making it perfect for use by trike users [6,7].

A thickness greater than 300 μm would make the material waterproof but not breathable, while a thickness below 150 μm might reduce strength and durability. Specific additives in the coating paste can improve performance and durability.

A combination with 10% binder (*Tubicoat PUT*), 2% thickener (*Tubicoat ASD 60*), 2% crosslinker (*Fixerer MF 200*), 3% softener (*Microsil 500*), and 4% water repellent (*Tubicoat A711*) provides excellent elasticity, waterproofing, and durability even after repeated washing.

3.2 Cushion

The main aspects we considered during the design of the cushion with independent air chambers were the number and layout of the cells, as well as the choice of suitable material. These decisions were made to balance [cost](#) and comfort with the hardware requirements of the microcontroller.

Another important aspect was ensuring the **stability of the cushion on the trike seat** during use. To achieve this, we designed a hooking system at the top part, aligned with the seat's fixing point, and added six Velcro strips: four placed on the sides of the cushion and two in the center. This setup ensures strong adhesion to the seat base and prevents unwanted movements.

3.2.1 Cells: number, shape, and layout

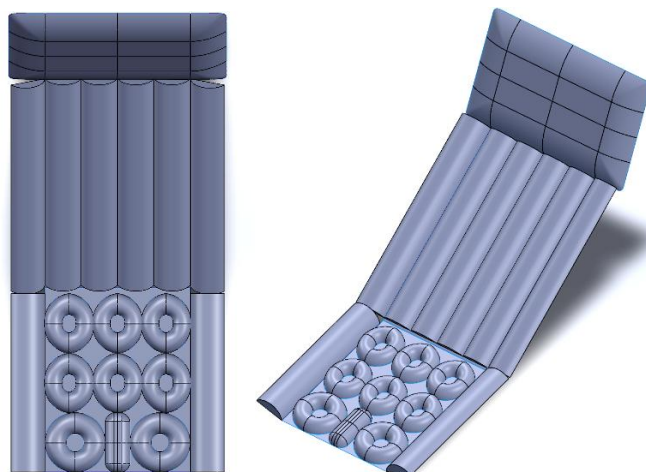


Figure 2 – CAD

Starting from a simple initial idea, inspired by the Flow Advance mattress, the project evolved into a more complex but functional solution. The entire design process was carried out using **SolidWorks**, which allowed us to experiment with different shapes and layouts of the air cells.

To create the cushion, the first step was to take the measurements of the trike in order to create a base for the seat and backrest. The initial dimensions planned for the different sections were:

- Seat: 31 x 36 cm
- Backrest: 40.5 x 36 cm
- Shoulder area: 22 x 36 cm
- Backrest inclination: 30° on the lateral plane relative to the z-axis
- Shoulder area inclination: 60° on the lateral plane relative to the z-axis

After that, we developed [several versions of the cushion](#), each with its own air cell configuration. The final version includes air cells of [different shapes and sizes](#) depending on their position, as shown in *Figure 2*.

On the **backrest**, we placed **six** vertically positioned semicylindrical **air cells**:

- two lateral ones, which are not dynamically controlled because they are inflated only at the beginning of the session to provide support and comfort;
- four central ones, connected to the valves, which will be inflated and deflated dynamically.

On the **seat**, we placed eleven **air cells**:

- two lateral, semicylindrical, vertically positioned cells, not dynamically controlled and inflated only at the beginning of the session;
- one rectangular cell placed between the thighs, static, inflated only initially to prevent the athlete from sliding off the cushion;
- six elliptical cells positioned in the sacral area, connected to the valves, which will be dynamically inflated and deflated;
- two circular-shaped cells in the area near the thighs, also dynamically controlled.

In the **shoulder area**, we decided to use a simple cushion without air cells, since in this area the risk of developing pressure sores is lower, but the presence of a soft cushion can still improve comfort without compromising functionality.

The lateral air chambers not dynamically controlled (both in the backrest and the seat) were slightly tilted to improve adaptation to the trike and provide better support to the user.

Regarding the semicylindrical shape, this was chosen based on the Flow Advance mattress. As for the seat part, which is more important in preventing pressure sores, we opted for circular and elliptical cushions cause they are better than other shapes in terms of comfort [8].

3.2.2 Material

Based on the description of the Flow Advance system, a high-quality anti-decubitus mattress, the material that best meets the functional and non-functional requirements is **polyurethane fiber (PU)** [9, 10, 11].

This polymer is highly elastic and capable of withstanding prolonged and repeated deformations without breaking or permanently deforming, an essential characteristic in a system that must continuously adapt to changes in body pressure even after repeated load cycles. PU is also biocompatible, an important factor even if the cushion is not in direct contact with the skin, to avoid possible long-term reactions.

Another advantage of PU is its compatibility with high-frequency welding and heat-sealing technologies, which allow for the creation of perfectly sealed air chambers, resistant to air leakage and moisture. These characteristics make polyurethane a highly reliable material for flexible and dynamic internal components.

As for the material thickness, a range between 0.2 mm and 0.5 mm is considered optimal. These values ensure good tensile and internal pressure resistance while maintaining the necessary flexibility to adapt to dynamic changes in body load.

Each cell is equipped with a small fitting or nozzle integrated directly into the chamber material, serving as a connection point between the chamber and the pneumatic tubing, which in turn allows connection to a 3/2 electropneumatic valve.

3.3 Pressure-Sensitive MAT

For detecting the pressure distribution between the body and the contact surface, the **Tekscan CONFORMat™** system in the Research configuration was chosen, a professional solution optimized for dynamic applications on complex geometries such as the trike saddle.

The system consists of:

- a flexible sensor mat based on a matrix of resistive sensors,
- an electronic acquisition module,
- a USB cable for data transmission,
- the Tekscan Research software and related SDK.

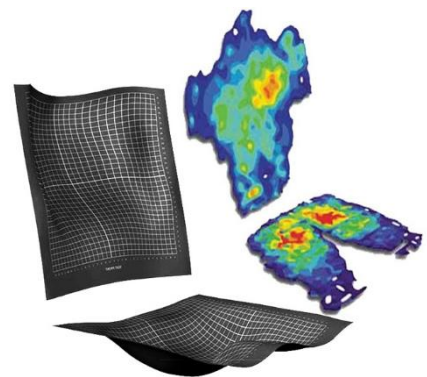


Figure 3 - Tekscan CONFORMat™

The CONFORMat™ mat features a matrix of about **1000 sensing points** distributed over a **50 x 50 cm** surface and offers a sampling frequency of up to **100 Hz**. In our case, given the trike dimensions, it was necessary to use **two pressure-sensitive mats** to fully cover the relevant surface areas (backrest and seat) and ensure accurate data detection.

The MAT allows excluding irrelevant portions by physically folding them or deactivating them via software. This optimizes data acquisition and processing only in areas of real interest, avoiding handling an excessive amount of redundant data. Because of this capability, it was decided not to use all 2000 sensors available in the two mats but to actively select only **672 sensors** based on the real geometry of the support surface and through a CAD analysis of the system. In particular:

- Backrest: each cell was associated with 100 sensors, for a total of 400 sensors distributed over 4 cells;
- Seat: the 6 rear cells are monitored by 32 sensors each, while the 2 larger front cells are controlled by 40 sensors each. In total, the seat uses 272 sensors.

In this way, the 672 active sensors were grouped into **12 logical areas**, each corresponding to one pneumatic cell.

The resistive sensors are thin (0.20 mm), flexible, and have an accuracy of $\pm 10\%$. Their structure allows perfect adaptation to deformable surfaces without altering comfort or introducing measurement distortions. Additionally, the sensors are replaceable and wired, offering robustness and continuity in sensing.

3.3.1 Integration with Embedded Systems

A crucial aspect of the project is the integration of the pressure acquisition system with an **Arduino microcontroller**, in order to perform real-time processing or to activate mechatronic feedback based on the collected data. The Tekscan Research software, while allowing data export in ASCII format, transmits data in real-time using a proprietary format (FSX), which is **not compatible** with embedded systems like Arduino.

To solve this issue, the use of the **Software Development Kit (SDK)** provided by *Tekscan* is essential. The SDK enables the development of custom applications in a Windows environment. Thanks to the SDK and its related **Application Programming Interfaces (APIs)**, it is possible to create a software bridge that converts the data into a format readable by Arduino, allowing communication between the BPMS system and the microcontroller via standard protocols.

3.4 Distributed Electropneumatic System

The regulation of airflow and its distribution inside the air chambers is managed by an electronic control system based on a microcontroller (Arduino), which operates according to data provided by the two pressure-sensitive MATs.

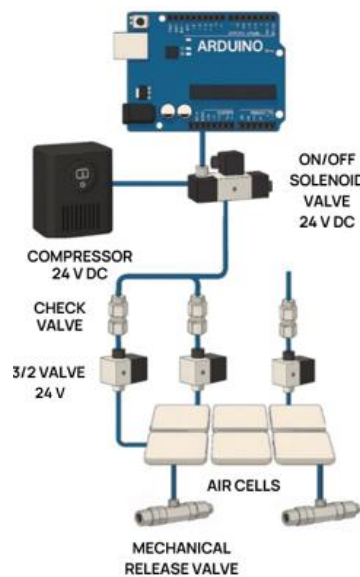


Figure 4 - Distributed Electropneumatic System structure

The microcontroller is directly connected, via a flexible hose, to a 24 V DC **compressor**, which acts as the main source of compressed air. This component was chosen for its compatibility with mobile applications, low noise level, and ease of electronic control. Its main technical features include:

- operating pressure sufficient to inflate all the cells simultaneously,
- airflow rate adequate to ensure fast response times,
- on-demand activation via a digital signal from the Arduino.

A suitable option meeting these requirements is the portable *CAFELE 24 V air compressor*, which provides a flow rate of 35 L/min and weighs about 308 g, making it very lightweight and suitable for mobile applications.

The compressor is connected to an **ON/OFF solenoid valve**, normally closed, which acts as the main pneumatic switch for the entire system. This component controls the airflow to the distribution network and is directly activated by the microcontroller. Its main features are:

- simple inflation control via digital signal,
- pneumatic isolation of the compressor when inactive,
- high sealing reliability.

A possible solution is the *Heschen Electric Pneumatic Valve 2V025-08 24VDC P*, weighing about 150 g.

From the ON/OFF valve, a network of **12 flexible tubes** branches out, one for each cell. Each line consists of the following functional sequence:

- a **check valve**, placed immediately downstream of the manifold, allows air to flow only toward the cell, preventing backflow and helping stabilize the internal pressure. Additionally, this valve acts as a passive safety release system, allowing air to escape if the pressure reaches a preset threshold, without the need for additional electronic control.

A possible technical solution is the *EFUTURETIME Pneumatic Check Valve* 10 mm, stainless steel quick-connect fitting, weighing about 30 g;

- a **3/2** normally closed **electropneumatic valve** receives air from the check valve and ensures its direction toward the chamber or toward the exhaust, allowing selective and controlled deflation. When inactive, the cell remains isolated, while when the valve is electrically activated, the chamber is connected to the exhaust. Its main features are:
 - individual pressure management for each cell,
 - partial exhaust to locally reduce pressure,
 - maintaining a minimum volume of air for postural stability.

A possible solution is the *Heschen Electric Pneumatic Valve*, 3V210-08, 24VDC, PT1/4", 2 positions 3 ways, weighing about 160 g.

Each air cell in the cushion also includes a **mechanical release valve** set at 0.2 MPa, which is not controlled by the microcontroller. Its purpose is to keep the internal pressure within safe limits. In case of accidental overpressure, due to overinflation or temperature changes, the valve automatically releases excess air to:

- protect the structure: preventing breakage or deformation of individual cells;
- safeguard the patient: reducing the risk of injury from excessive pressure or discomfort;
- ensure therapeutic effectiveness: maintaining optimal conditions for pressure sore prevention and comfort.

A possible solution is the *Aurorad 1/2" BSP Male Thread 2/3/4/5/6/7/8/9/10/12.5/15 Bar Brass Air Safety Valve, Vent Valve, Pressure Release Valve for Air Compressor (Size: 2 Bar)*, weighing about 12 g.

Regarding power supply, the entire pneumatic system is powered by an external power source, sized to support simultaneous activation of the compressor and multiple valves. The microcontroller's power can be managed in two ways: through a separate line or by tapping from the main power source using a voltage regulator.

Finally, the microcontroller manages the entire monitoring and control process in real time: it continuously collects data from the pressure-sensitive MAT and, based on the detected loads, activates the individual 3/2 valves to selectively deflate areas under excessive pressure. In addition to this dynamic control, a periodic automatic reinflation cycle is implemented, during which the compressor and ON/OFF valve are reactivated to restore optimal pressure in all cells, ensuring comfort and functionality over the long term.

3.5 Microcontroller

The **Arduino Mega 2560 microcontroller** was chosen because it has many digital pins and more memory compared to other models in the same family. This microcontroller works as the main control unit for the whole system, connecting both to the sensor system (MAT) and the pneumatic system (valves and compressor).



Figure 5 - Arduino Mega 2560 microcontroller

The microcontroller reads pressure data from the pressure-sensitive MAT through a USB connection. A processing speed of **10 Hz** was chosen to fit the device's computing limits and keep it working well.

The microcontroller also controls turning the electropneumatic system on and off, using 12 digital output pins to control each 3/2 electropneumatic valve separately, one pin for the main ON/OFF valve, and one more pin for the compressor.

Because valves and the compressor work at 24 V and need more current than the microcontroller can give, we use relay circuits to connect them. Each **relay** is controlled by one Arduino output pin and has a **flyback diode** to protect the microcontroller from voltage spikes caused by inductive currents. This setup makes the control of the system's high-power parts safe and reliable.

3.5.1 Software

The software installed on the microcontroller was **developed in C++ using the Arduino IDE development environment**. Its task is to manage in real time the inflation and deflation operations of the 12 air cells based on the pressure data detected by the two MATs. The sensors in the matrix transmit data at a frequency of 100 Hz; however, due to the microcontroller's limited computational capacity, processing occurs at 10 Hz. Every 100 ms, a new matrix (the latest complete data packet received) is processed and then used to update the system.

During the initial setup phase, some estimates were necessary since exact measurements of the volume and internal pressure of the air cells were not available. The total volume of the cells was estimated to be **5 liters** (backrest = $40 \times 36 \times 2$ + seat = $36 \times 30 \times 2 = 5040 \text{ cm}^3 = 5.040 \text{ dm}^3 = 5 \text{ L}$) of air at atmospheric pressure. Considering a **desired pressure of 0.2 MPa** (2 times atmospheric pressure), the volume to be introduced corresponds approximately to 10 liters. Since the compressor used has a **flow rate of 35 L/min**, the expected time for fully inflating the cells until the release valves open is about **17.14 seconds**.

Pressure control is performed on each individual sensor and not on the average of the values within the cell. This approach is motivated by the fact that highly concentrated pressure areas, which are clinically relevant, could be missed by average control. Each sensor in the pressure-sensitive matrix covers an area of about 2.5 cm^2 , making point control an adequate solution to precisely identify risk zones.

The [software](#) performs a set of essential functions:

- it receives, via USB connection, **data for 672 pressure values** coming from the two sensor mats;
- **every 100 milliseconds**, it saves and analyzes the latest complete row of received data;
- it groups these data into **12 logical sets**, each corresponding to one of the pneumatic cells;
- it calculates the **average pressure for each cell**, useful for an overall assessment;
- for each sensor, it keeps track of the **pressure-time** product value, following the **clinical curve by Reswick and Rogers** [12], and triggers automatic actions if the critical threshold of 300 mmHg-hours is exceeded.

At system startup, all the cells are inflated for 20 seconds to reach the pressure needed to open the release valves, ensuring the full operation of the cushion before the patient uses it. During normal operation, the system constantly monitors the pressure applied by the patient's body. If one or more sensors detect prolonged pressure above the threshold, the associated cell is selectively deflated to reduce the risk of pressure sores. Additionally, every three minutes the system performs a short general inflation of all the cells lasting three seconds, to compensate for any small leaks and maintain balanced pressure across the different sections of the circuit.

4. Main challenges

During the design and development of the system, several technical problems appeared that affect how well the device works, how comfortable it is, and how well it fits in real situations.

- **Shape and number of air chambers:** A future version could have air cells with different, irregular shapes made for each person, and more cells, to improve comfort and fit. This would allow better and more exact pressure control, but it would also make the system more complex, bigger, and more expensive.
- **Wiring:** The system needs many electrical and air tubes connections, causing problems with space, weight, and mess. To fix this, we suggest making holes in the trike frame to pass the air tubes under the frame. A net under the frame could hold the tubes and valves neatly, making the system lighter, safer, and better organized.
- **Software interface:** Right now, it is necessary to make software using the *Tekscan* SDK to change the data so Arduino can read it.
- **Outdoor use and total weight:** The total weight of the device makes it hard to use in sports or outside. We need to make lighter versions.

5. Conclusion

The project led to the development of an innovative system for dynamic pressure management, designed for athletes with spinal cord injuries using trikes for FES-cycling. The system integrates a cushion with independent air chambers, two *Tekscan CONFORMat™* pressure mats, and an Arduino Mega 2560 microcontroller, allowing **real-time monitoring of pressure distribution and selective regulation of individual pneumatic cells**.

This strategy enables the reduction of load in critical areas, effectively preventing the onset of pressure ulcers and improving comfort during physical activity. The CAD design enabled precise integration of the system with the trike structure, ensuring functionality, stability, and customization. The multidisciplinary approach combined biomechanics, electronics, and design, resulting in an adaptive, intelligent, and user-friendly solution.

Although the prototype demonstrated technical robustness and strong application potential, some aspects still need to be optimized, such as overall weight, power management, and cable organization. Nevertheless, this work represents a concrete step toward the adoption of advanced assistive technologies in the sports and rehabilitation fields, offering athletes with spinal cord injuries **greater safety, comfort, and freedom in sports practice**.

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7. Attachment

7.1 CAD

7.1.1 Final CAD dimensions

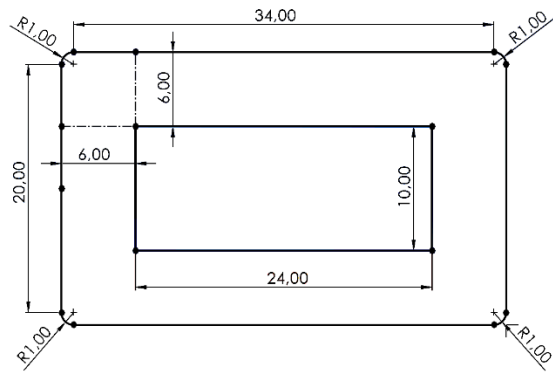


Figure 6 - Top plane relative to shoulder area

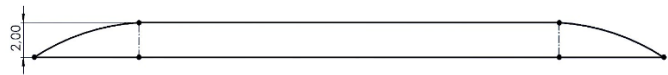


Figure 7 - Front plane relative to shoulder area

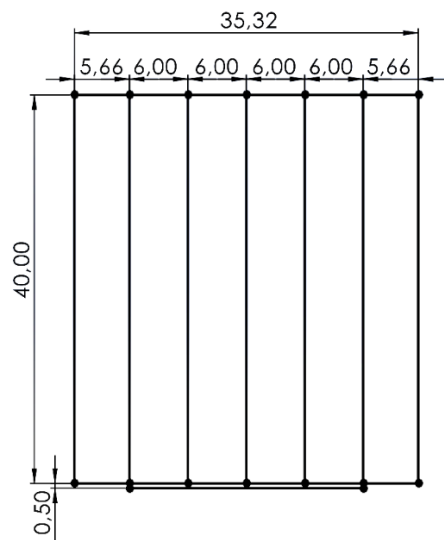


Figure 8 – Top plane relative to backrest



Figure 9 - Front plane relative to backrest

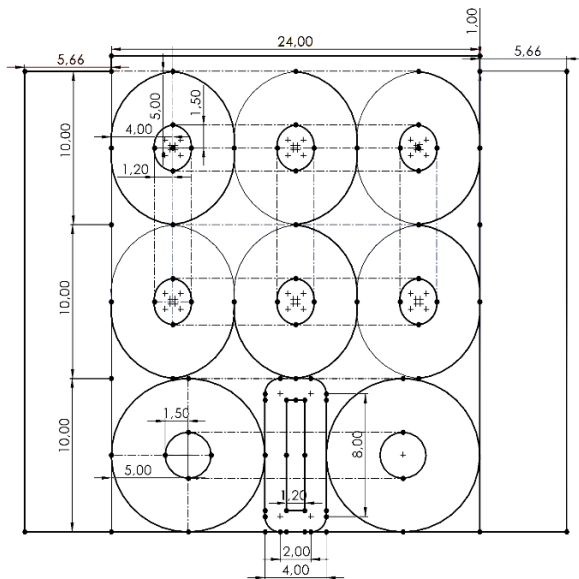


Figure 10 - Top plane relative to seat

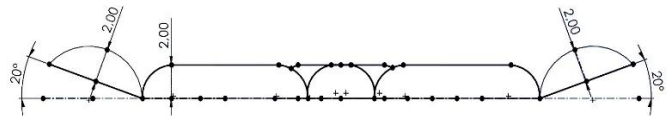


Figure 11 - Front plane relative to seat

7.1.2 Alternative CAD Designs

The design process of the cushion began with the creation of an **initial CAD model** (figure 12) inspired by a commercial anti-decubitus mattress, with the aim of replicating a simple and functional configuration. This first version served as the conceptual foundation of the project.

Subsequently, the design evolved into a **second version** (figure 13), which introduced structural improvements to enhance comfort and optimize pressure redistribution.

At this stage, a **matrix-based configuration** (figure 14) was developed to allow more precise control of pressure zones and facilitate future integration of sensors and active systems.

The following version (figure 15) maintained the matrix layout but introduced **circular air cells** in the sacral area, based on literature evidence indicating their greater effectiveness in terms of comfort and pressure injury prevention.

The final CAD model combines the strengths of all previous versions, resulting in a solution optimized in terms of geometry, functionality, and user safety.

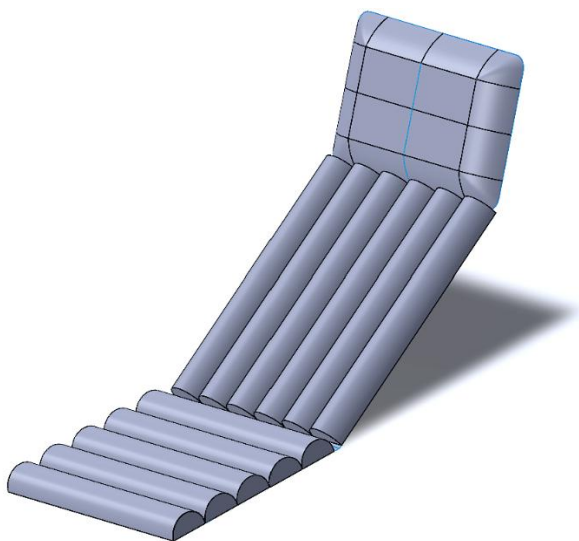


Figure 12 – CAD 1

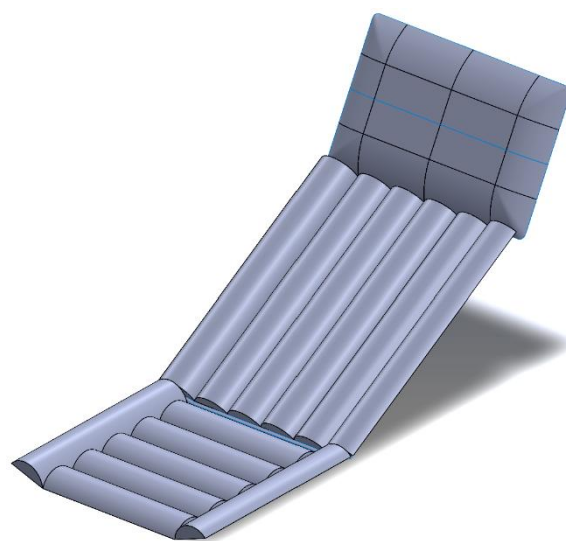


Figure 13 - CAD 2

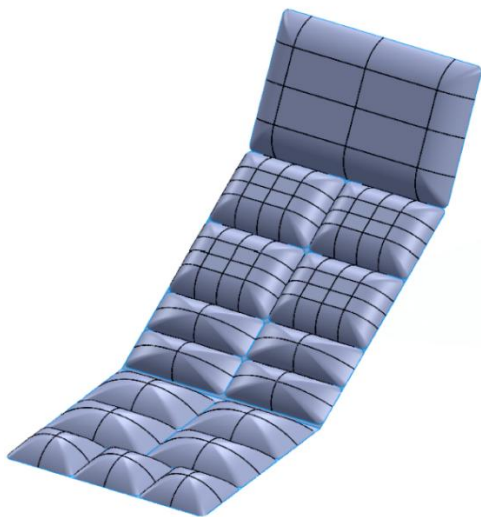


Figure 14 - CAD 3

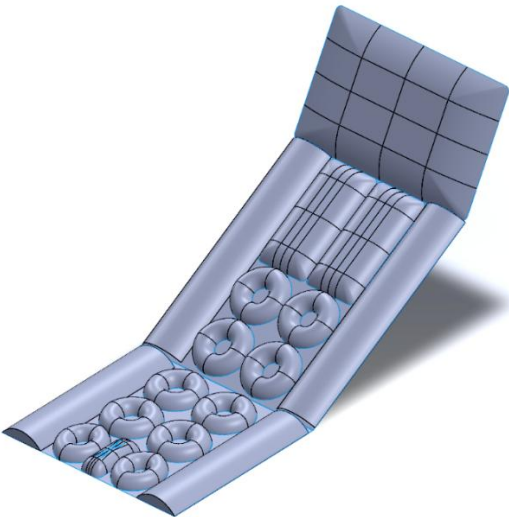


Figure 15 - CAD 4

7.2 Sensor Selection

Table 1 - Sensors analyzed

SENSOR TYPE	OPERATING PRINCIPLE	UNIT COST	ADVANTAGES	DISADVANTAGES	FITS OUR PROJECT	APPLICATIONS IN LITERATURE
FSR (RESISTIVE)	Resistance varies with applied force	5–8 €	- Low cost - Flexible - Easy to integrate with Arduino	- Non-linear response - Affected by temperature and humidity - Mechanical wear	Excellent for wide coverage at low cost	Yang et al. (2019); low-cost rehabilitation seating, ergonomic prototypes
CAPACITIVE	Capacitance varies between two electrodes	15–50 €	- Linear and precise response - Good stability - Minimal external influence	- Require advanced readout circuitry - More expensive	Good for sensitive areas (ischium, lumbar area)	Misaki et al. (2021); Chang & Huang (2014); active air anti-decubitus mattresses
PIEZO-ELECTRIC	Generates electric charge in response to dynamic changes	>50 €	- High sensitivity to dynamic changes - Good real-time response	-- Not suitable for static pressure - Complex conditioning circuitry	Low for use on static cushions	Kistler (2020) – Dynamic Load and Gait Detection Using Piezoelectric Force Plates.

FIBER OPTIC (FBG)	Optical index changes with deformation	>300 €	- Extremely precise	- Expensive	Not suitable for prototyping or limited-budget projects	Lee et al. (2019) – Active Body Pressure Relief System with ToF Optical Pressure Sensors (adaptive mattress with ToF sensors)
			- Immune to interference - Sensor networks on one fiber	- Require specialized equipment - Complex integration		

7.3 Code

```
#include <math.h>
#define NUM_SENSORI_TOTALI 672
#define NUM_SENSORI_SCHIENALE 400
#define NUM_SENSORI_SEDUTA 272
#define NUM_CELLE_SCHIENALE 4
#define NUM_CELLE_SEDUTA 8
#define NUM_CELLE_TOTALI (NUM_CELLE_SCHIENALE + NUM_CELLE_SEDUTA)
#define SENSORI_SEDUTA_STANDARD 32
#define SENSORI_SEDUTA_SPECIALI 40

#define COMPRESSOR_PIN 2
#define VALVOLA_GENERALE_PIN 3
#define CONTROL_INTERVAL 100

typedef struct {
    int pinSgonfia;
    int numSensori;
    int indiceInizio;
} Cella;

Cella celleSchienale[NUM_CELLE_SCHIENALE] = {
    {4, 100, 0},
    {5, 200, 100},
    {6, 300, 200},
    {7, 400, 300}
};

Cella celleSeduta[NUM_CELLE_SEDUTA] = {
    {8, 32, 0},
    {9, 64, 32},
    {10, 96, 64},
    {11, 128, 96},
    {12, 160, 128},
    {13, 192, 160},
    {A0, 232, 192},
    {A1, 272, 232}
};

int16_t valoriRaw[NUM_SENSORI_TOTALI];
float valoriLogici[NUM_CELLE_TOTALI];
uint16_t pressioneTempo[NUM_SENSORI_TOTALI];

unsigned long lastUpdate = 0;
```

```
unsigned long lastGonfiaggioTotale = 0;

const float sogliaBassa = 10.0;
const float limitePT = 300.0;
const unsigned long intervalloGonfiaggioTotale = 180000;
const unsigned long durataGonfiaggioTotale = 3000;

String inputBuffer = "";

void setup() {
    Serial.begin(115200);

    pinMode(COMPRESSOR_PIN, OUTPUT);
    pinMode(VALVOLA_GENERALE_PIN, OUTPUT);
    digitalWrite(COMPRESSOR_PIN, LOW);
    digitalWrite(VALVOLA_GENERALE_PIN, LOW);

    for (int i = 0; i < NUM_CELLE_SCHIENALE; i++) {
        pinMode(celleSchienale[i].pinSgonfia, OUTPUT);
        digitalWrite(celleSchienale[i].pinSgonfia, LOW);
    }

    for (int i = 0; i < NUM_CELLE_SEDUTA; i++) {
        pinMode(celleSeduta[i].pinSgonfia, OUTPUT);
        digitalWrite(celleSeduta[i].pinSgonfia, LOW);
    }

    digitalWrite(COMPRESSOR_PIN, HIGH);
    digitalWrite(VALVOLA_GENERALE_PIN, HIGH);
    delay(20000);
    digitalWrite(COMPRESSOR_PIN, LOW);
    digitalWrite(VALVOLA_GENERALE_PIN, LOW);

    lastUpdate = millis();
    lastGonfiaggioTotale = millis();
}

void loop() {
    while (Serial.available()) {
        char c = Serial.read();
        if (c == '\n') {
            parseData(inputBuffer);
            inputBuffer = "";
        } else {
            inputBuffer += c;
        }
    }

    unsigned long now = millis();
    if (now - lastUpdate >= CONTROL_INTERVAL) {
        aggiornaValoriLogici();
        aggiornaPressioneTempo();
        aggiornaControllo();
        lastUpdate = now;
    }

    if (now - lastGonfiaggioTotale >= intervalloGonfiaggioTotale) {
```

```

        gonfiaTutteLeCelle();
        lastGonfiaggioTotale = now;
    }
}

void parseData(String input) {
    int index = 0;
    char* token = strtok((char*)input.c_str(), ",");
    while (token != NULL && index < NUM_SENSORI_TOTALI) {
        valoriRaw[index] = atoi(token);
        token = strtok(NULL, ",");
        index++;
    }
}

void aggiornaValoriLogici() {
    for (int i = 0; i < NUM_CELLE_SCHIENALE; i++) {
        long somma = 0;
        for (int j = 0; j < celleSchienale[i].numSensori; j++) {
            somma += valoriRaw[celleSchienale[i].indiceInizio + j];
        }
        valoriLogici[i] = somma / (float) celleSchienale[i].numSensori;
    }

    for (int i = 0; i < NUM_CELLE_SEDUTA; i++) {
        long somma = 0;
        for (int j = 0; j < celleSeduta[i].numSensori; j++) {
            somma += valoriRaw[NUM_SENSORI_SCHIENALE + celleSeduta[i].indiceInizio
+ j];
        }
        valoriLogici[i + NUM_CELLE_SCHIENALE] = somma / (float)
celleSeduta[i].numSensori;
    }
}

void aggiornaPressioneTempo() {
    float deltaOre = CONTROL_INTERVAL / 3600000.0;
    for (int i = 0; i < NUM_SENSORI_TOTALI; i++) {
        float pressione = valoriRaw[i];
        float valoreDaAggiungere = pressione * deltaOre;
        uint16_t incremento = (uint16_t) ceil(valoreDaAggiungere);
        pressioneTempo[i] += incremento;
    }
}

void aggiornaControllo() {
    // Schienale
    for (int i = 0; i < NUM_CELLE_SCHIENALE; i++) {
        digitalWrite(celleSchienale[i].pinSgonfia, LOW);
        bool allarme = false;
        for (int j = 0; j < celleSchienale[i].numSensori; j++) {
            int idx = celleSchienale[i].indiceInizio + j;
            if (pressioneTempo[idx] > limitePT) {
                allarme = true;
                pressioneTempo[idx] = 0;
            }
        }
        if (allarme) {

```

```
        digitalWrite(celleSchienale[i].pinSgonfia, HIGH);
    }
}

// Seduta
for (int i = 0; i < NUM_CELLE_SEDUTA; i++) {
    digitalWrite(celleSeduta[i].pinSgonfia, LOW);
    bool allarme = false;
    for (int j = 0; j < celleSeduta[i].numSensori; j++) {
        int idx = NUM_SENSORI_SCHIENALE + celleSeduta[i].indiceInizio + j;
        if (pressioneTempo[idx] > limitePT) {
            allarme = true;
            pressioneTempo[idx] = 0;
        }
    }
    if (allarme) {
        digitalWrite(celleSeduta[i].pinSgonfia, HIGH);
    }
}

void gonfiaTutteLeCelle() {
    for (int i = 0; i < NUM_CELLE_SCHIENALE; i++) {
        digitalWrite(celleSchienale[i].pinSgonfia, LOW);
    }
    for (int i = 0; i < NUM_CELLE_SEDUTA; i++) {
        digitalWrite(celleSeduta[i].pinSgonfia, LOW);
    }

    digitalWrite(COMPRESSOR_PIN, HIGH);
    digitalWrite(VALVOLA_GENERALE_PIN, HIGH);
    delay(durataGonfiaggioTotale);
    digitalWrite(COMPRESSOR_PIN, LOW);
    digitalWrite(VALVOLA_GENERALE_PIN, LOW);
}
```

7.4 Overall cost estimate

Table 2 - Indicative project cost

COMPONENT	COMMERCIAL EXAMPLE	APPROXIMATE PRICE (EUR)
AIR CELL CUSHION	Custom or OEM (e.g., Roho, Vicair)	~3,000–5,000 €
BREATHABLE COVER	Micro-perforated PU cover	~200–300 €
MATS	Tekscan CONFORMat system	~13,000 €
COMPRESSOR (24 V DC)	CAFELE 24 V air compressor	~38 €
ON/OFF VALVE (2/2)	Heschen 2V025-08 24 V DC	~12 €
3/2 VALVES (12 UNITS)	Heschen 3V210-08 24 V DC	~12 € each
CHECK VALVES (12 UNITS)	EFUTURETIME 10 mm quick-connect	~5 € each
SAFETY VALVES (12 UNITS)	Aurorad 2 bar ½"	~6 € each
RELAY BOARDS (2 UNITS)	ARCELI 8-channel 5 V	~5 € each
COMPRESSOR RELAY	CGEAMDY 24 V 30 A automotive relay	~12 €
WIRING AND PROTECTION	16 AWG cables, fuses, 1N4007 diodes	~10–15 € total

8. Report of the focus group with end-users

With the aim of gathering useful insights for the design of a device, a **focus group** (a qualitative research technique used to gather opinions, perceptions, attitudes, and ideas through guided discussion) was organized with the aim of collecting useful insights for the design of a device. It was conducted with three SCI athletes from Politecnico di Milano who are experienced in using the FES-bike.

To prepare for this session, each student group involved developed a **set of questions** divided by topics (performance and comfort), in order to guide the conversation toward key issues without influencing the participants' responses. **Two moderators** were selected to lead the discussion and define the methods of data collection: notes and audio recording.

The two moderators explained the importance of the session to the participants. The questions asked were intentionally broad to give them the opportunity to speak freely, leading to a conversation that allowed us to identify some of the strengths and critical issues of the trike. During the meeting, we listened to the athletes' direct experiences and collected numerous observations about the difficulties they encounter during training and competitions. This moment of exchange served as **the starting point for defining the functional requirements of the system we designed**.

The topics the participants focused on the most were insufficient comfort, especially during prolonged training sessions, and the difficulty of using the trike outdoors, due to limitations related to transport, safety, and the ability to personalize the vehicle. There was also strong interest in solutions that could make training more complete, such as integrating exercises for the upper limbs, and that would allow autonomous and safe outdoor use of the trike, even in non-competitive contexts. The **risk of developing pressure ulcers** was a key concern shared by all participants, as it is considered a significant obstacle not only to consistent training but also to general well-being.

This aspect was one of the fundamental inputs during the preliminary analysis phase for defining the system's functional requirements.